Machine Learning Models for Predicting Agricultural Yields in Developing Countries

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Abstract

This research is aimed at creating strong machine learning models to forecast crop yields in different states of India and overcoming the fact that the traditional forecasting methods are not receptive to the nonexistent linear relationships between environmental, agronomic, and temporal factors. The study has used a 19,689 record of 1997 to 2020 data, including characteristics of crop type, area, production, rainfall, fertilizer use, pesticide work, and seasonality. There was a systematic workflow, i.e., data preprocessing, outlier credit, categorical coding, scaling of features, and separation of features and targets. Linear regression, random forest, and XGBoost models had been trained and compared, and a model based on LSTM time series had been designed to learn time-dependent metrics. Measures that were used to evaluate the models included MAE, RMSE, R^2, confusion matrices, and ROC-AUC curves. Findings showed that ensemble methods performed better compared to the linear and sequential models, whereby Random Forest had an R^2 of 0.9741, whilst XGBoost had 0.9709, indicating the ability to capture typical nonlinear trends. The performance of LSTM was apparent to be affected (R^2 = -0.0158). The feature importance analysis revealed that the key factors predicting the yield are production, area, and rainfall. The research also comes to the conclusion that machine learning, especially ensemble methods, is accurate, explainable, and scalable for crop yield prediction to enhance informed agricultural planning and optimization of resources and improve food security policies.

Acknowledgement

We have been grateful to all people and institutions that helped us undertake this study successfully. Academic supervisors deserve special mention in this regard because their coaching, support, and critical feedback were invaluable and supportive. Appreciation is also offered to colleagues and peers who were supportive during the process of data collection, data analysis, and implementation of the model. Their input has been very valuable to this research on its quality and rigor.

Declaration

This dissertation is presented in submission of the degree requirements and depicts original work done independently. Every source, data, and thought used out of available literature has been duly credited. This entire work has not been presented in other academic institutions to receive academic credit or an award.

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Chapter 1: Introduction

In developing countries, agriculture is a key strength towards economic stability and food supply. To counter its significance, crop production is highly erratic because of swift climatic changes, irregular rain patterns, soil erosion, and lack of technological application in the agricultural systems (Kuradusenge et al., 2023). Experience-based judgments are the common practice by farmers and are prone to errors in the situations where the environmental patterns are not within historical records. Most governments also do not have powerful forecasting systems, which limit them in their capacity to design timely interventions, allocate resources effectively, and solve arising food shortages (Gu et al., 2022). Subsequently, the unpredictability's of farm production pose a long-term threat to the local markets, supply chain, and national food stock.

Machine learning has also become an influential tool in the discovery of hidden patterns of complex and multidimensional agricultural data. ML models are capable of making predictions that are more accurate than traditional statistics, as they involve the use of climate records, soil parameters, and past yield value data. These models have been in a position to coordinate nonlinear relationship between the environmental factors and thereby render it more fitting and precise in future forecasting. This type of predictive potential can offer the climatic, agricultural, and long-range food security that might be necessary in other locations that are already constrained by economic and infrastructural factors.Therefore, it has become highly important to work out valid ML-driven yield prediction models in crop planning today.

1.1 Rationale of the Study

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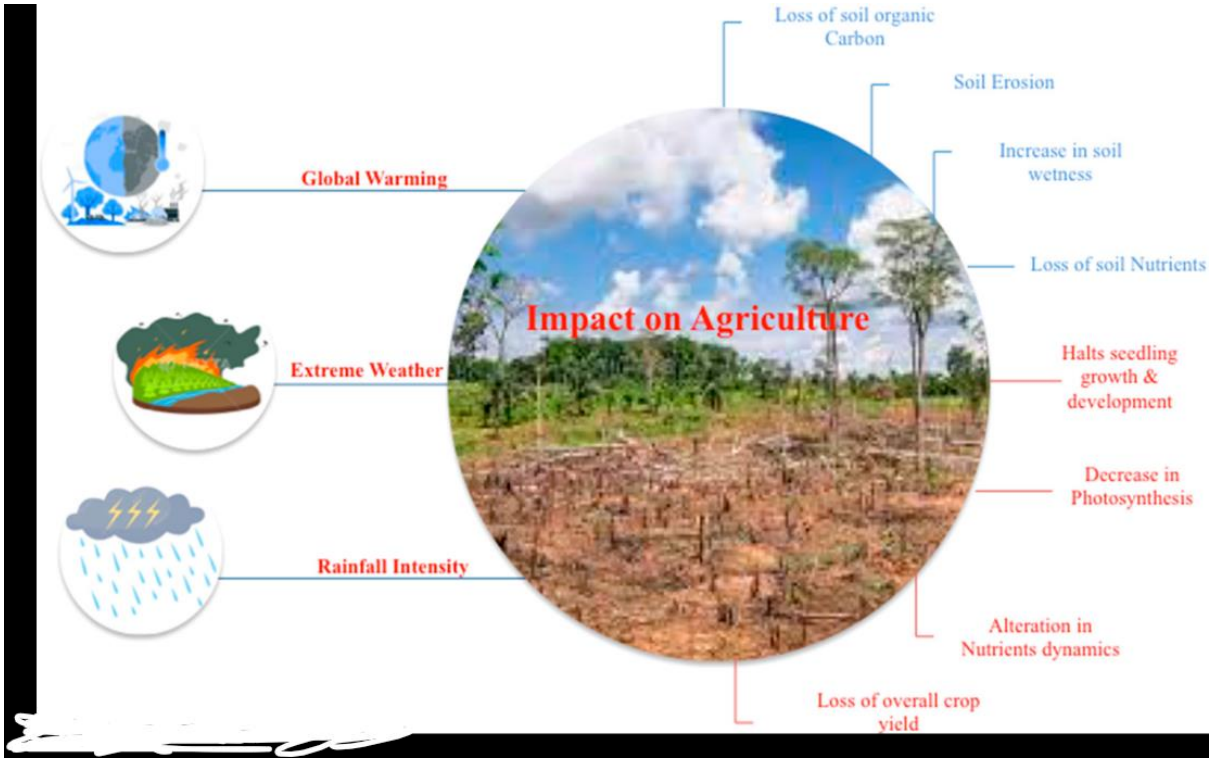

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Factor Category	Key Influences	Typical Effects on Crop Yield	Description
Climate Factors	Rainfall patterns	Yield fluctuations	Directly affect soil moisture availability and plant growth
Temperature Conditions	Heat stress or cold stress	Plant growth disruption	Impacts photosynthesis and flowering stages
Extreme Weather	Droughts, floods	Reduced yield consistency	Causes sudden changes in the production cycle
Seasonal Variations	Monsoon variability	Uncertain crop planning	Affects sowing decisions and harvest timing

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Soil / Agronomic Element	Influence on Crops	Observed Impact	Explanation
Soil Fertility	Nutrient support	Enhanced crop health	Determines nutrient availability for plant development
Soil Texture	Water retention	Variability in root growth	Controls drainage and moisture balance
Organic Matter	Soil enrichment	Better soil structure	Improves soil aeration and microbial activity
Farming Practices	Crop rotation, irrigation	Yield stability	Shapes long-term soil productivity

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Approach Type	Strengths	Limitations	Suitability for Yield Forecasting
Traditional Statistical Models	Easy to interpret	Struggle with nonlinear relationships	Suitable only for stable and consistent environments
Linear ML Models	Simple implementation	Limited adaptation to dynamic variables	Useful for baseline predictions
Tree-Based ML Models	Handle complex interactions	Require tuning	Strong for mixed climate–soil datasets
Deep Learning Models	Capture temporal patterns	High computational needs	Ideal for time-series and sequential data

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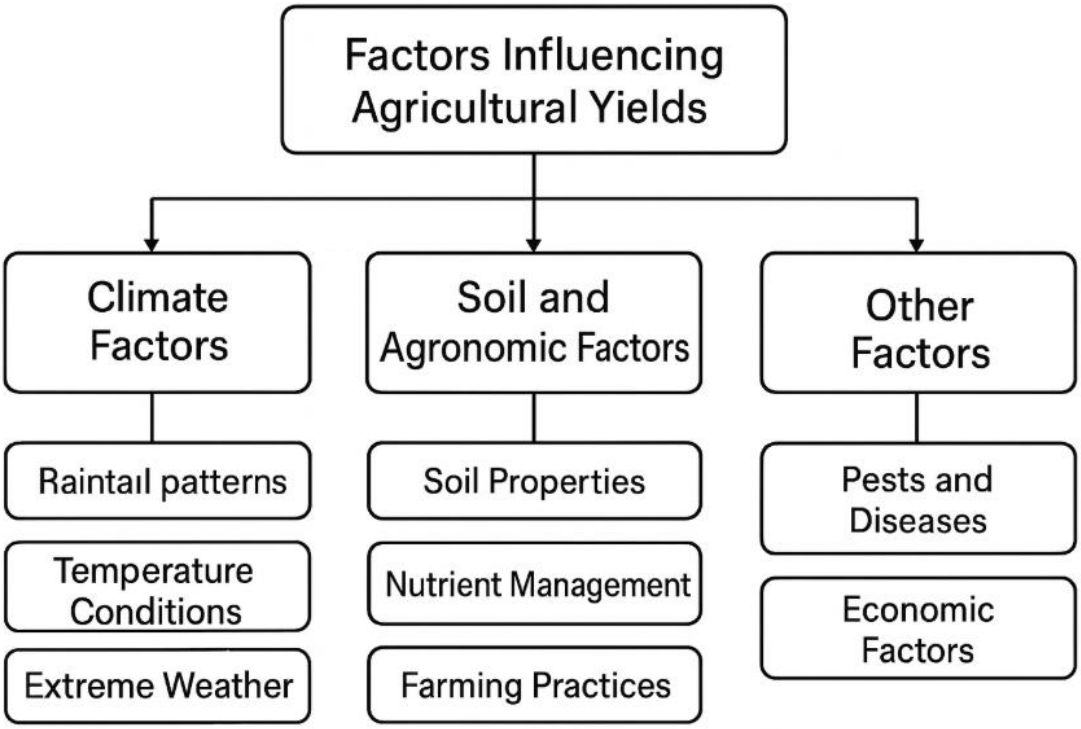

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Factor Category	Data Included	Importance in Prediction	Reason for Inclusion
Environmental Factors	Temperature, rainfall	High relevance	Directly influence crop growth stages
Soil Properties	Texture, fertility	Medium to high relevance	Affect nutrient availability and water balance
Agronomic Practices	Planting method, irrigation	Context-specific usefulness	Shape productivity patterns
Climatic Patterns	Seasonal shifts, weather trends	Critical relevance	Capture unpredictable conditions

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```
Dataset URL: https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset
License(s): CC-BY-SA-4.0
Downloading crop-yield-in-indian-states-dataset.zip to crop_yield_indian_states_dataset
  0% 0.00/476k [00:00<?, ?B/s]
100% 476k/476k [00:00<00:00, 522MB/s]
Archive:  crop_yield_indian_states_dataset/crop-yield-in-indian-states-dataset.zip
  inflating: crop_yield_indian_states_dataset/crop_yield.csv
['crop-yield-in-indian-states-dataset.zip', 'crop_yield.csv']
```

```
Dataset Shape: (19689, 10)

Column Names:
Index(['Crop', 'Crop_Year', 'Season', 'State', 'Area', 'Production',
       'Annual_Rainfall', 'Fertilizer', 'Pesticide', 'Yield'],
      dtype='object')
```

	Crop	Crop_Year	Season	State	Area	Production	Annual_Rainfall	Fertilizer	Pesticide	Yield
0	Arecanut	1997	Whole Year	Assam	73814.0	56708	2051.4	7024878.38	22882.34	0.796087
1	Arhar/Tur	1997	Kharif	Assam	6637.0	4685	2051.4	631643.29	2057.47	0.710435
2	Castor seed	1997	Kharif	Assam	796.0	22	2051.4	75755.32	246.76	0.238333
3	Coconut	1997	Whole Year	Assam	19656.0	126905000	2051.4	1870661.52	6093.36	5238.051739
4	Cotton(lint)	1997	Kharif	Assam	1739.0	794	2051.4	165500.63	539.09	0.420909

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	Crop_Year	Area	Production	Annual_Rainfall	Fertilizer	Pesticide	Yield
count	19689.000000	1.968900e+04	1.968900e+04	19689.000000	1.968900e+04	1.968900e+04	19689.000000
mean	2009.127584	1.799266e+05	1.643594e+07	1437.755177	2.410331e+07	4.884835e+04	79.954009
std	6.498099	7.328287e+05	2.630568e+08	816.909589	9.494600e+07	2.132874e+05	878.306193
min	1997.000000	5.000000e-01	0.000000e+00	301.300000	5.417000e+01	9.000000e-02	0.000000
25%	2004.000000	1.390000e+03	1.393000e+03	940.700000	1.880146e+05	3.567000e+02	0.600000
50%	2010.000000	9.317000e+03	1.380400e+04	1247.600000	1.234957e+06	2.421900e+03	1.030000
75%	2015.000000	7.511200e+04	1.227180e+05	1643.700000	1.000385e+07	2.004170e+04	2.388889
max	2020.000000	5.080810e+07	6.326000e+09	6552.700000	4.835407e+09	1.575051e+07	21105.000000

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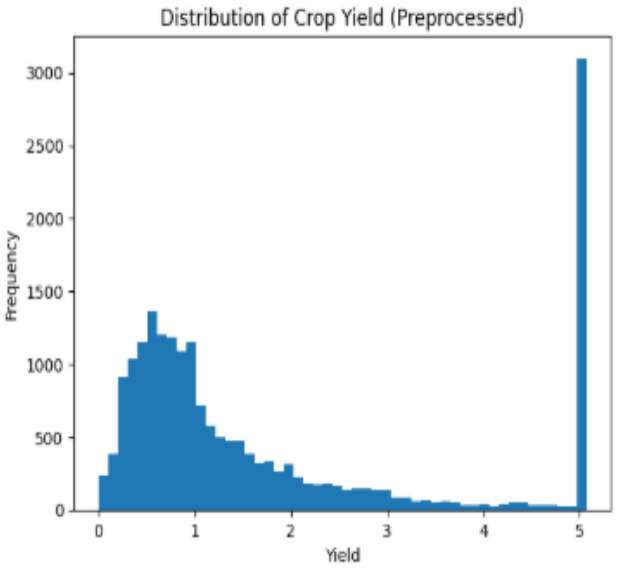
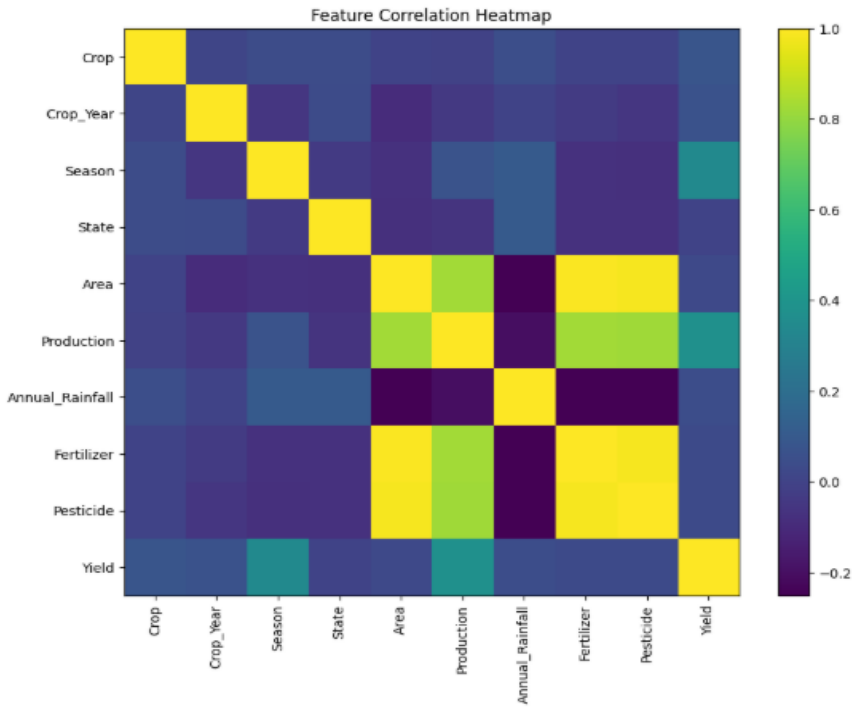
	Area	Production	Annual_Rainfall	Fertilizer	Pesticide	Yield
count	19689.000000	19689.000000	19689.000000	1.968900e+04	19689.000000	19689.000000
mean	49091.865530	80716.405988	1371.231397	6.528931e+06	13013.902888	1.783845
std	68864.272117	115591.715658	614.694947	9.148706e+06	18328.408492	1.652179
min	0.500000	0.000000	301.300000	5.417000e+01	0.090000	0.000000
25%	1390.000000	1393.000000	940.700000	1.880146e+05	356.700000	0.600000
50%	9317.000000	13804.000000	1247.600000	1.234957e+06	2421.900000	1.030000
75%	75112.000000	122718.000000	1643.700000	1.000385e+07	20041.700000	2.388889
max	185695.000000	304705.500000	2698.200000	2.472760e+07	49569.200000	5.072222

	Crop	Crop_Year	Season	State	Area	Production	Annual_Rainfall	Fertilizer	Pesticide	Yield
0	0	1997	4	2	0.359007	-0.207705	1.106542	0.054211	0.538437	0.796087
1	1	1997	1	2	-0.616516	-0.657775	1.106542	-0.644620	-0.597799	0.710435
2	8	1997	1	2	-0.701337	-0.698116	1.106542	-0.705383	-0.696594	0.238333
3	9	1997	4	2	-0.427458	1.937810	1.106542	-0.509185	-0.377595	5.072222
4	11	1997	1	2	-0.687643	-0.691437	1.106542	-0.695573	-0.680644	0.420909

```
X_train shape: (15751, 9)
X_test shape : (3938, 9)
y_train shape: (15751,)
y_test shape : (3938,)
```





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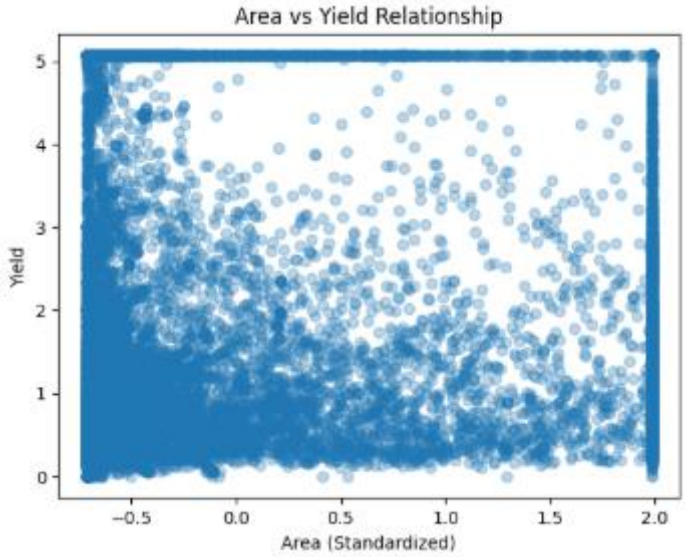

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```python
# Install Kaggle
!pip install -q kaggle

KAGGLE_USERNAME = "your_kaggle_username"
KAGGLE_KEY = "your_kaggle_api_key"

import os

# Set Kaggle API credentials
os.environ["KAGGLE_USERNAME"] = KAGGLE_USERNAME
os.environ["KAGGLE_KEY"] = KAGGLE_KEY

# Directory to save the dataset
dataset_dir = "crop_yield_indian_states_dataset"
os.makedirs(dataset_dir, exist_ok=True)

# Download the dataset from Kaggle
!kaggle datasets download -d akshatgupta7/crop-yield-in-indian-states-dataset -p {dataset_dir}

# Unzip the dataset if it is zipped
!unzip -o {dataset_dir}/*.zip -d {dataset_dir}

# List downloaded files
os.listdir(dataset_dir)
```

Preprocessing of Dataset

```python
#  3: Handle missing values and remove duplicates

# Drop duplicate rows
df = df.drop_duplicates()

# Confirm no missing values remain
df.isnull().sum()

# Confirm updated shape
df.shape
```

```python
#  4: Outlier detection and treatment using IQR method (numerical columns)

import numpy as np

# Select numerical columns
numerical_cols = [
    'Area', 'Production', 'Annual_Rainfall',
    'Fertilizer', 'Pesticide', 'Yield'
]

# Apply IQR-based outlier capping
for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    df[col] = np.where(df[col] < lower_bound, lower_bound, df[col])
    df[col] = np.where(df[col] > upper_bound, upper_bound, df[col])

# Verify updated statistics after outlier handling
df[numerical_cols].describe()
```

```python
# 5: Encode categorical variables and scale numerical features

from sklearn.preprocessing import LabelEncoder, StandardScaler

# Copy dataframe
df_processed = df.copy()

# Encode categorical columns
categorical_cols = ['Crop', 'Season', 'State']
label_encoders = {}

for col in categorical_cols:
    le = LabelEncoder()
    df_processed[col] = le.fit_transform(df_processed[col])
    label_encoders[col] = le

# Scale numerical columns
numerical_cols = [
    'Area', 'Production', 'Annual_Rainfall',
    'Fertilizer', 'Pesticide'
]

scaler = StandardScaler()
df_processed[numerical_cols] = scaler.fit_transform(df_processed[numerical_cols])

# Verify processed dataset
df_processed.head()
```

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Train-test split + visual insights from preprocessed dataset

```python
#  6: Train-test split + visual insights from preprocessed dataset

import matplotlib.pyplot as plt
import pandas as pd
from sklearn.model_selection import train_test_split

# Define features and target
X = df_processed.drop(columns=['Yield'])
y = df_processed['Yield']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    shuffle=True
)

# Print split shapes (output)
print("X_train shape:", X_train.shape)
print("X_test shape :", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape :", y_test.shape)

#  VISUAL INSIGHTS

# 1. Distribution of target variable (Yield)
plt.figure()
plt.hist(df_processed['Yield'], bins=50)
plt.title("Distribution of Crop Yield (Preprocessed)")
plt.xlabel("Yield")
plt.ylabel("Frequency")
plt.show()

# 2. Correlation heatmap (numerical features)
corr = df_processed.corr()

plt.figure(figsize=(10, 8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Feature Correlation Heatmap")
plt.show()
```

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```python
# 3. Yield vs Area relationship
plt.figure()
plt.scatter(df_processed['Area'], df_processed['Yield'], alpha=0.3)
plt.xlabel("Area (Standardized)")
plt.ylabel("Yield")
plt.title("Area vs Yield Relationship")
plt.show()
```

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LINEAR REGRESSION – Training, Evaluation, and Plots

```python
# 7: LINEAR REGRESSION - Training, Evaluation, and Plots

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score, confusion_matrix, roc_curve, auc
import numpy as np
import matplotlib.pyplot as plt

# Train model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

# Predictions
y_pred_lr = lr_model.predict(X_test)

# Regression metrics
lr_mae = mean_absolute_error(y_test, y_pred_lr)
lr_rmse = np.sqrt(mean_squared_error(y_test, y_pred_lr))
lr_r2 = r2_score(y_test, y_pred_lr)

print("Linear Regression MAE:", lr_mae)
print("Linear Regression RMSE:", lr_rmse)
print("Linear Regression R2:", lr_r2)

# Binary conversion for classification-style plots
threshold = y_test.median()
y_test_bin = (y_test >= threshold).astype(int)
y_pred_bin = (y_pred_lr >= threshold).astype(int)

# Confusion Matrix
cm = confusion_matrix(y_test_bin, y_pred_bin)
plt.figure()
plt.imshow(cm)
plt.title("Linear Regression Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# ROC Curve
fpr, tpr, _ = roc_curve(y_test_bin, y_pred_lr)
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, label="AUC = %.2f" % roc_auc)
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Linear Regression ROC Curve")
plt.legend()
plt.show()
```

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RANDOM FOREST – Training, Evaluation

```python
# 8: RANDOM FOREST – Training, Evaluation, and Visual Plots

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score, confusion_matrix, roc_curve, auc
import numpy as np
import matplotlib.pyplot as plt

# Train Random Forest model
rf_model = RandomForestRegressor(
    n_estimators=200,
    max_depth=15,
    random_state=42,
    n_jobs=-1
)

rf_model.fit(X_train, y_train)

# Predictions
y_pred_rf = rf_model.predict(X_test)

# Evaluation metrics
rf_mae = mean_absolute_error(y_test, y_pred_rf)
rf_rmse = np.sqrt(mean_squared_error(y_test, y_pred_rf))
rf_r2 = r2_score(y_test, y_pred_rf)

print("Random Forest MAE:", rf_mae)
print("Random Forest RMSE:", rf_rmse)
print("Random Forest R2:", rf_r2)

# Classification-style plots

# Binary threshold (median yield)
threshold = y_test.median()
y_test_bin = (y_test >= threshold).astype(int)
y_pred_bin_rf = (y_pred_rf >= threshold).astype(int)

# Confusion Matrix
cm_rf = confusion_matrix(y_test_bin, y_pred_bin_rf)

plt.figure()
plt.imshow(cm_rf)
plt.title("Random Forest Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

[redacted]

```python
# ROC Curve
fpr_rf, tpr_rf, _ = roc_curve(y_test_bin, y_pred_rf)
roc_auc_rf = auc(fpr_rf, tpr_rf)

plt.figure()
plt.plot(fpr_rf, tpr_rf, label="AUC = %.2f" % roc_auc_rf)
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Random Forest ROC Curve")
plt.legend()
plt.show()

#  Feature Importance

importances = rf_model.feature_importances_

plt.figure(figsize=(10, 5))
plt.bar(X.columns, importances)
plt.xticks(rotation=90)
plt.title("Random Forest Feature Importance")
plt.show()
```

[redacted]

XGBOOST – Training, Evaluation

```python
# 9: XGBOOST – Training, Evaluation, and Visual Plots

!pip install -q xgboost

from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score, confusion_matrix, roc_curve, auc
import numpy as np
import matplotlib.pyplot as plt

# Train XGBoost model
xgb_model = XGBRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    objective='reg:squarederror'
)

xgb_model.fit(X_train, y_train)

# Predictions
y_pred_xgb = xgb_model.predict(X_test)

# Evaluation metrics
xgb_mae = mean_absolute_error(y_test, y_pred_xgb)
xgb_rmse = np.sqrt(mean_squared_error(y_test, y_pred_xgb))
xgb_r2 = r2_score(y_test, y_pred_xgb)

print("XGBoost MAE:", xgb_mae)
print("XGBoost RMSE:", xgb_rmse)
print("XGBoost R2:", xgb_r2)

# Classification-style plots

# Binary threshold (median yield)
threshold = y_test.median()
y_test_bin = (y_test >= threshold).astype(int)
y_pred_bin_xgb = (y_pred_xgb >= threshold).astype(int)

# Confusion Matrix
cm_xgb = confusion_matrix(y_test_bin, y_pred_bin_xgb)
```

[redacted]

```python
plt.figure()
plt.imshow(cm_xgb)
plt.title("XGBoost Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# ROC Curve
fpr_xgb, tpr_xgb, _ = roc_curve(y_test_bin, y_pred_xgb)
roc_auc_xgb = auc(fpr_xgb, tpr_xgb)

plt.figure()
plt.plot(fpr_xgb, tpr_xgb, label="AUC = %.2f" % roc_auc_xgb)
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("XGBoost ROC Curve")
plt.legend()
plt.show()

# Feature Importance

importances_xgb = xgb_model.feature_importances_

plt.figure(figsize=(10, 5))
plt.bar(X.columns, importances_xgb)
plt.xticks(rotation=90)
plt.title("XGBoost Feature Importance")
plt.show()
```

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[redacted]

[redacted]

LSTM – Time-Series Model Training, Evaluation

```python
# 10: LSTM – Time-Series Model Training, Evaluation, and Plots

import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

# Sort data by time
df_lstm = df_processed.sort_values("Crop_Year")

# Select features for LSTM
lstm_features = ['Area', 'Production', 'Annual_Rainfall', 'Fertilizer', 'Pesticide']

# Scale features
scaler = MinMaxScaler()
scaled_features = scaler.fit_transform(df_lstm[lstm_features])

# Create time-series sequences
def create_sequences(X, y, time_s=3):
    Xs, ys = [], []
    for i in range(len(X) - time_s):
        Xs.append(X[i:i + time_s])
        ys.append(y.iloc[i + time_s])
    return np.array(Xs), np.array(ys)

X_seq, y_seq = create_sequences(scaled_features, df_lstm['Yield'])

# Train-test split (time-aware)
split_index = int(0.8 * len(X_seq))
X_train_lstm, X_test_lstm = X_seq[:split_index], X_seq[split_index:]
y_train_lstm, y_test_lstm = y_seq[:split_index], y_seq[split_index:]

# Build LSTM model
lstm_model = Sequential()
lstm_model.add(LSTM(64, activation='relu', input_shape=(X_train_lstm.shape[1], X_train_lstm.shape[2])))
lstm_model.add(Dense(1))

lstm_model.compile(optimizer='adam', loss='mse')

# Train model
history = lstm_model.fit(
    X_train_lstm,
    y_train_lstm,
    epochs=25,
    batch_size=32,
    validation_data=(X_test_lstm, y_test_lstm),
    verbose=1
)

# Predictions
y_pred_lstm = lstm_model.predict(X_test_lstm).flatten()
```

[redacted]

```python
# Evaluation metrics
lstm_mae = mean_absolute_error(y_test_lstm, y_pred_lstm)
lstm_rmse = np.sqrt(mean_squared_error(y_test_lstm, y_pred_lstm))
lstm_r2 = r2_score(y_test_lstm, y_pred_lstm)

print("LSTM MAE:", lstm_mae)
print("LSTM RMSE:", lstm_rmse)
print("LSTM R2:", lstm_r2)

#  Training history plot

plt.figure()
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("LSTM Training and Validation Loss")
plt.legend()
plt.show()

#  Actual vs Predicted plot

plt.figure()
plt.plot(y_test_lstm[:100], label='Actual')
plt.plot(y_pred_lstm[:100], label='Predicted')
plt.xlabel("Time s")
plt.ylabel("Yield")
plt.title("LSTM Actual vs Predicted Yield")
plt.legend()
plt.show()
```

[redacted]

[redacted]

Final Model Comparison

```python
# 11: Final Model Comparison - Table and Performance Plots

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Create comparison table
results_df = pd.DataFrame({
    "Model": [
        "Linear Regression",
        "Random Forest",
        "XGBoost",
        "LSTM"
    ],
    "MAE": [
        lr_mae,
        rf_mae,
        xgb_mae,
        lstm_mae
    ],
    "RMSE": [
        lr_rmse,
        rf_rmse,
        xgb_rmse,
        lstm_rmse
    ],
    "R2 Score": [
        lr_r2,
        rf_r2,
        xgb_r2,
        lstm_r2
    ]
})

print(results_df)

# MAE comparison
plt.figure()
plt.bar(results_df["Model"], results_df["MAE"])
plt.title("Model Comparison - MAE")
plt.ylabel("MAE")
plt.xticks(rotation=20)
plt.show()

# RMSE comparison
plt.figure()
plt.bar(results_df["Model"], results_df["RMSE"])
plt.title("Model Comparison - RMSE")
plt.ylabel("RMSE")
plt.xticks(rotation=20)
plt.show()
```

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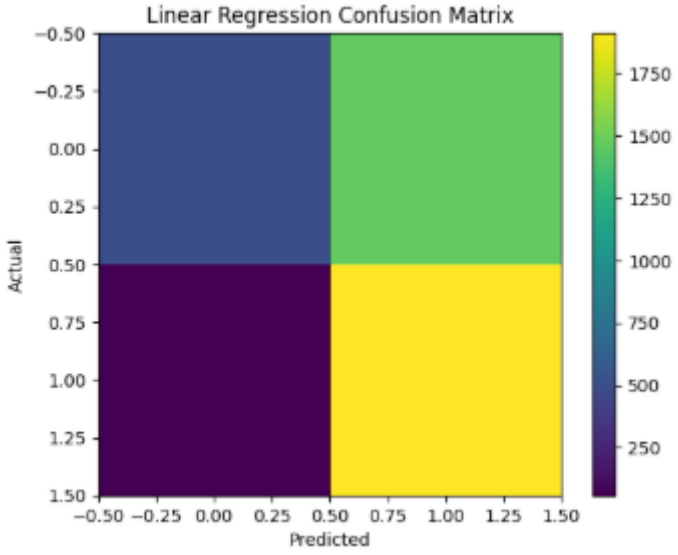
[redacted]

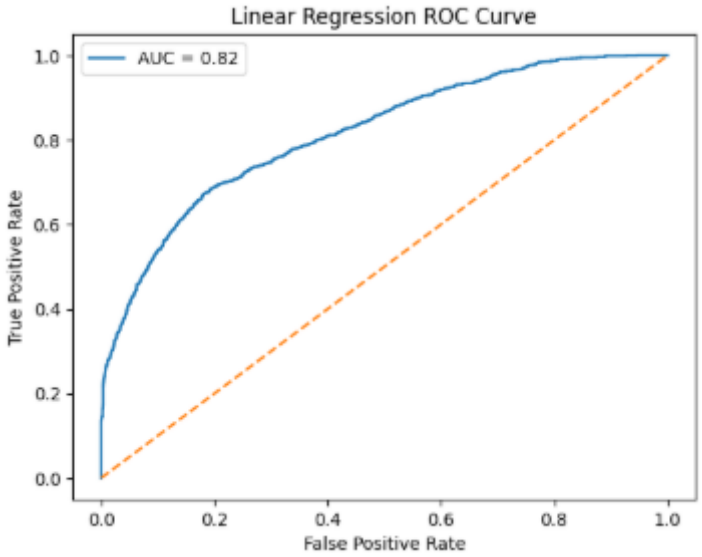
[redacted]

[redacted]

[redacted]



[redacted]

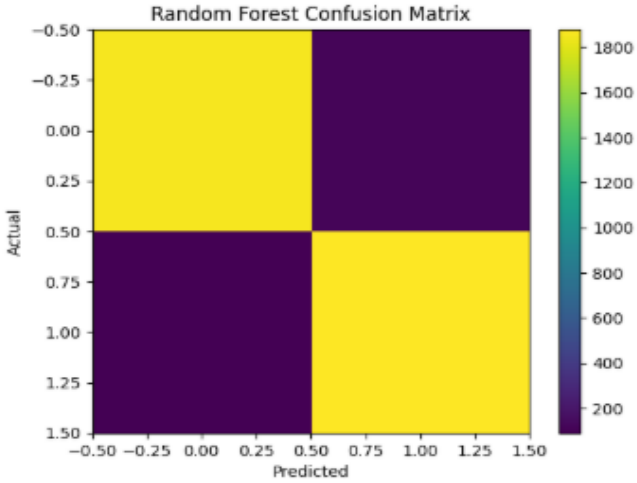


[redacted]

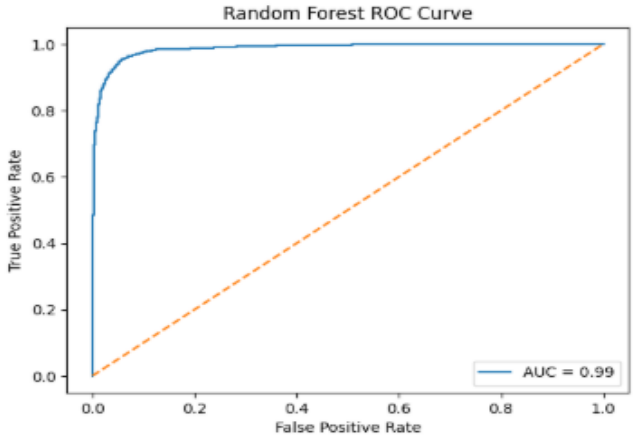
[redacted]

[redacted]

[redacted]



[redacted]

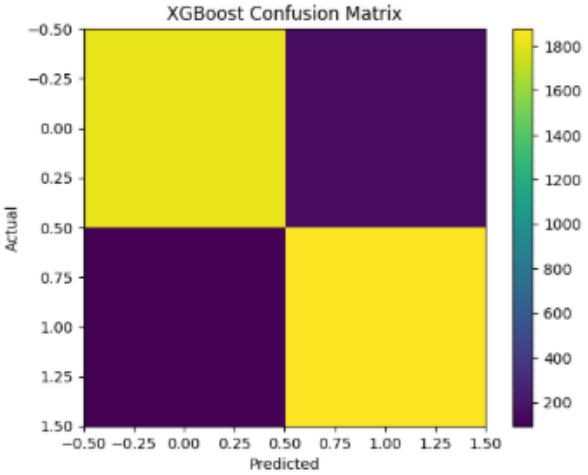


[redacted]

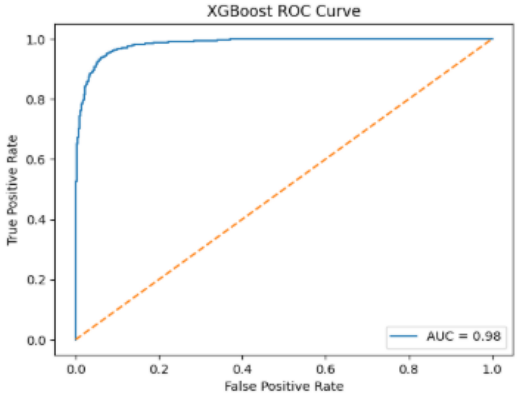
[redacted]

[redacted]

[redacted]



[redacted]



[redacted]

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[redacted]

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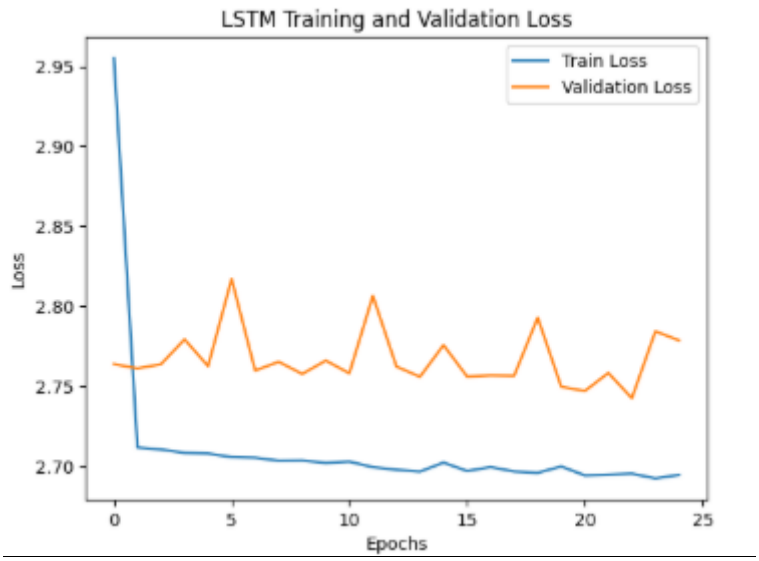
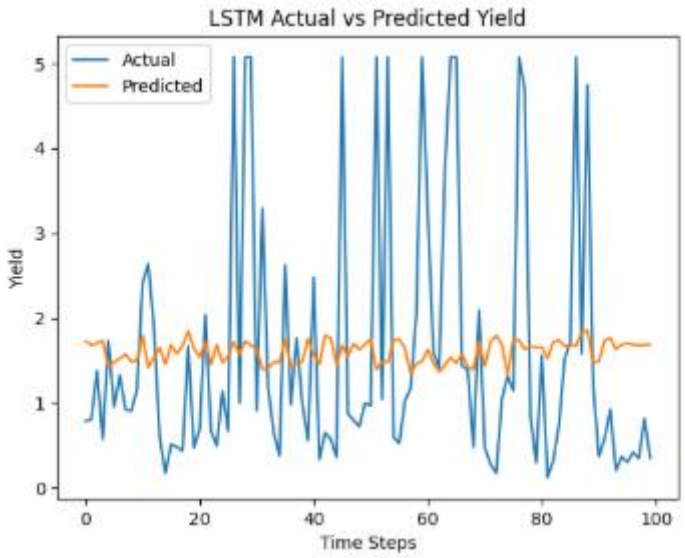
```
Epoch 1/25
/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an `input_shape`/`input_dim` argumen
  super().__init__(**kwargs)
493/493 ━━━━━━━━━━━━━━━━━━━━ 56 5ms/step - loss: 3.4930 - val_loss: 2.7637
Epoch 2/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7353 - val_loss: 2.7610
Epoch 3/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7124 - val_loss: 2.7636
Epoch 4/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7028 - val_loss: 2.7792
Epoch 5/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6656 - val_loss: 2.7625
Epoch 6/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7013 - val_loss: 2.8170
Epoch 7/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 2.6936 - val_loss: 2.7597
Epoch 8/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7343 - val_loss: 2.7652
Epoch 9/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6856 - val_loss: 2.7575
Epoch 10/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7107 - val_loss: 2.7659
Epoch 11/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7140 - val_loss: 2.7579
Epoch 12/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7057 - val_loss: 2.8064
Epoch 13/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 2.6628 - val_loss: 2.7622
Epoch 14/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6705 - val_loss: 2.7558
Epoch 15/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7106 - val_loss: 2.7756
Epoch 16/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6740 - val_loss: 2.7559
Epoch 17/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7185 - val_loss: 2.7567
Epoch 18/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6836 - val_loss: 2.7565
Epoch 19/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 2.6379 - val_loss: 2.7928
Epoch 20/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7305 - val_loss: 2.7495
Epoch 21/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7047 - val_loss: 2.7469
Epoch 22/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6884 - val_loss: 2.7582
Epoch 23/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.6676 - val_loss: 2.7424
Epoch 24/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - loss: 2.7324 - val_loss: 2.7842
Epoch 25/25
493/493 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 2.6980 - val_loss: 2.7786
124/124 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
LSTM MAE: 1.3130087384742768
LSTM RMSE: 1.6669018776118482
LSTM R2: -0.01583602418939711
```

[redacted]



[redacted]



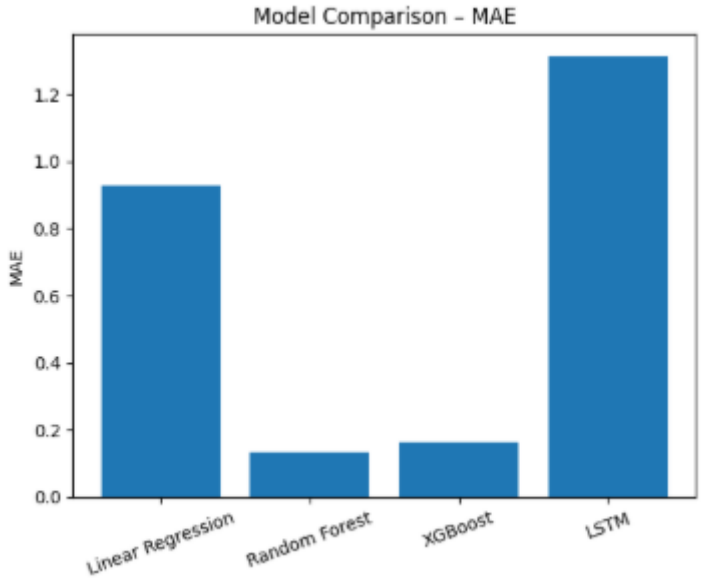
[redacted]

[redacted]

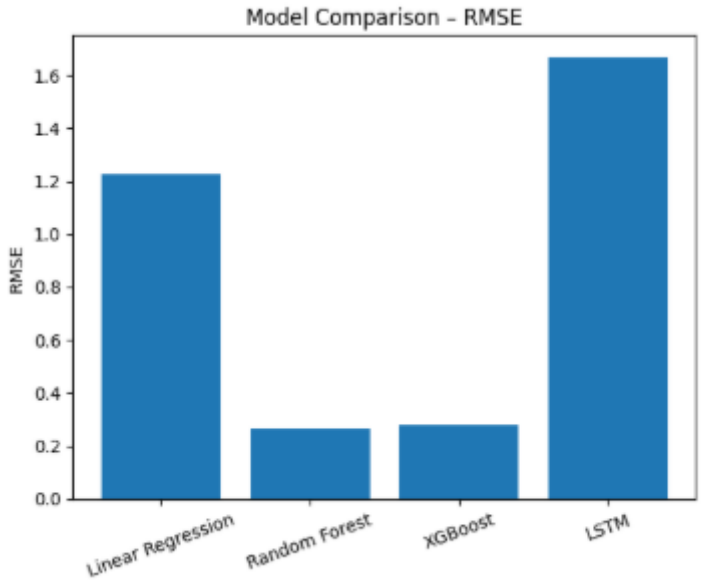
[redacted]

	Model	MAE	RMSE	R2 Score
0	Linear Regression	0.927753	1.225677	0.444102
1	Random Forest	0.132324	0.264475	0.974117
2	XGBoost	0.165339	0.280669	0.970851
3	LSTM	1.313009	1.666902	-0.015836

[redacted]



[redacted]



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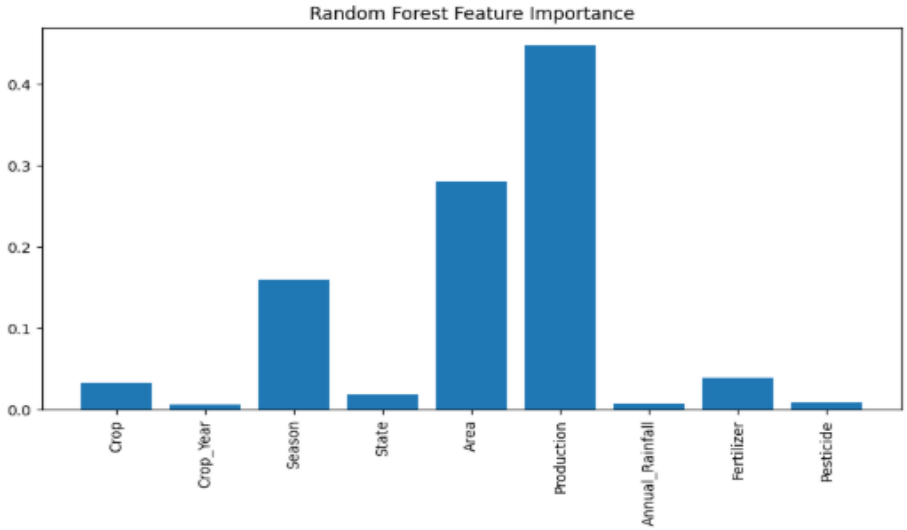
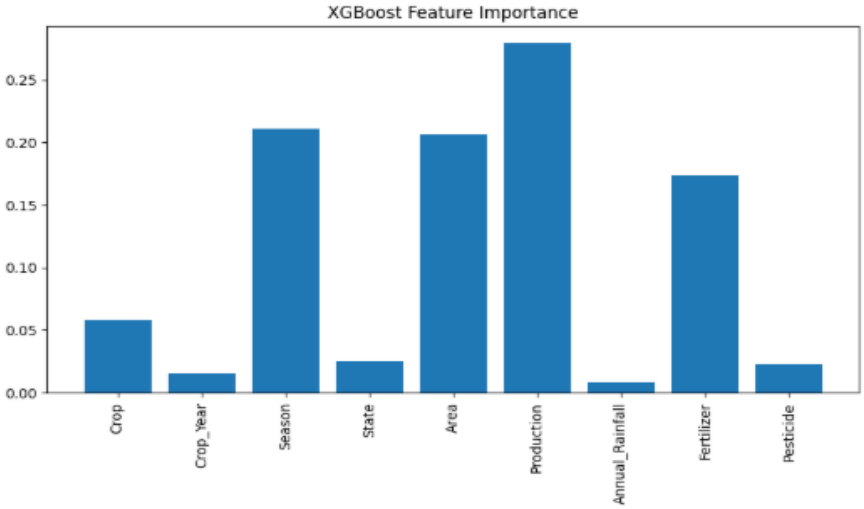

Random Forest Feature Importance
0.4
0.3
0.2
0.1
0.0
Crop
Crop_Year
Season
State
Area
Production
Annual_Rainfall
Fertilizer
Pesticide


XGBoost Feature Importance
0.25
0.20
0.15
0.10
0.05
0.00
Crop
Crop_Year
Season
State
Area
Production
Annual_Rainfall
Fertilizer
Pesticide

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